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Interface Circuits for a Six-Phase Variable Reluctance Energy Harvester

A comparative study of rectification and power conditioning
topologies

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Why energy harvesting?



- Wireless sensor networks and IoT devices have grown very fast, and are widely used to **collect data** that can then be processed for many different purposes.
- **Millions of small nodes** therefore need to be powered.
- Batteries are expensive to maintain, have a finite life, and in many embedded or rotating installations **cannot be replaced at all**; they are also a poor choice for the environment.

Energy harvesting

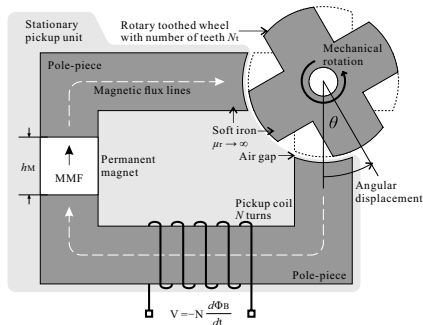
Convert energy already available in the environment into electrical power, so that the node powers itself for its whole lifetime.

This thesis targets **rotating motion**, converted through **Variable Reluctance Energy Harvesting (VREH)**.

The variable reluctance principle



- Magnet and coil stay **completely still**.
- Only a toothed wheel rotates across the air gap.
- The passing teeth modulate the reluctance, hence the flux linked by the coil: an EMF is induced.



Advantages

- No moving electrical parts.
- Easily scaled to different shaft sizes.
- Very small counter torque.

Stationary pickup unit (magnet, pole piece, coil) facing the rotating toothed wheel.

From single-unit prototypes for railway monitoring to an optimised **six-phase** design for large vehicles (buses): six pickup units, electrically shifted by 60° , mounted around the shaft.

rpm	f (Hz)	V_{Lpk} (V)	R_{coil} (Ω)	L_{coil} (mH)	P_o (W)
100	36.7	0.91	6	21.68	0.41
200	73.3	1.35	6	21.46	0.91
300	110.0	1.59	6	21.27	1.26
400	146.7	1.74	6	21.12	1.51

Target operating range: **100–400 rpm**, i.e. 20–80 km/h for a typical large commercial vehicle. Each phase delivers less than 2 V peak: the rectifier choice becomes critical.

- More phases mean more power, but a much harder electronics problem: **six AC signals**, all shifted in time, all to be rectified.
- Previous work compared rectifier topologies for a **single-phase** VREH only.
- How to condition the power of a six-phase harvester has never been studied systematically.

Research question

Which interface circuit extracts the most power from the six-phase MP-VREH, across its entire 100–400 rpm speed range?

A complete path, from theory to a working prototype:

1. **Model** the harvester: electrical model of one coil, extended to the six-phase system.
2. **Propose** rectifier topologies able to combine six phases.
3. **Simulate** each candidate in MATLAB/Simulink, sweeping the compensation capacitor and the load.
4. **Design and build** a single reconfigurable prototype board.
5. **Measure** every topology on the same hardware, on the same test bench.
6. **Assess** compatibility with commercial power management ICs.

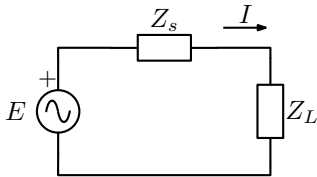
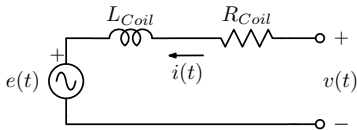
Coil model and matching



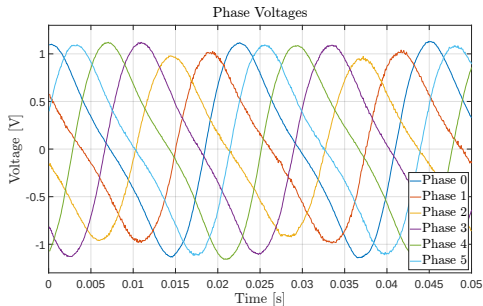
- Each coil: ideal source in series with R_{Coil} and L_{Coil} .
- At low speed a purely **resistive** load is enough to extract the maximum power.
- As speed (and frequency) grows, $\omega_e L_{Coil}$ can no longer be neglected: the matched-load condition is lost.
- A **series compensation capacitor** C_m cancels the inductive reactance and restores it:

$$C_m = \frac{1}{\omega_e^2 L_{Coil}}$$

valid at one frequency at a time.

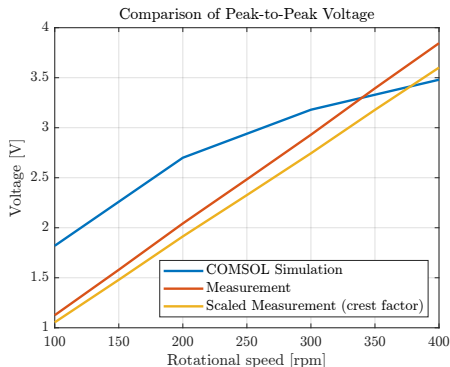


The six real phase voltages



- Measured open-circuit voltage of the six coils at 100 rpm.
- The 60° electrical shift between phases is clearly visible.
- The waveform is **not** a clean sine wave, and the six units are not perfectly identical.

- The RMS voltage of each coil was computed as a function of speed.
- Each source was then replaced by an **equivalent sinusoid**, scaled through the measured crest factor.
- It delivers the same power as the real waveform, so the simulation stays realistic while remaining tractable.



FEM prediction, raw measured peak and crest-factor corrected value.

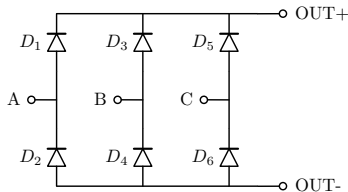
Strategy 1: FWR bridges



The six phases split into **two three-phase groups** ($k = 0, 2, 4$ and $k = 1, 3, 5$), each feeding a standard three-phase bridge.

- **Star (Y)**: line voltage is $\sqrt{3}$ times the phase EMF, so the fixed two-diode drop costs relatively less.
- **Delta (Δ)**: line voltage equals the coil EMF, no amplitude gain, but no floating node.

Expectation: **higher open-circuit voltage for the star connection**. Both are carried forward as candidates.



Three-phase bridge for one group; the current path always crosses two diodes.

Strategy 2: NVC bridges

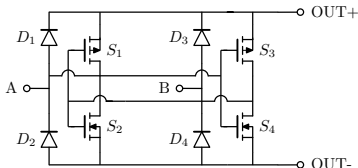


The six phases split into **three two-phase groups**, each pairing phases 180° apart ($k = 0, 3; 1, 4; 2, 5$) in **anti-series**:

$$v_A - v_B = e_k - e_{k+3} = 2 e_k$$

ideally double the voltage, at the cost of a doubled series impedance.

- Four MOSFETs with **cross-coupled gates**: two conduct at a time, no gate driver needed.
- They bypass the parallel Schottky diodes, replacing the fixed drop with a small R_{on} ; below $V_{off} \approx 1.2$ V the bridge falls back to diode behaviour.

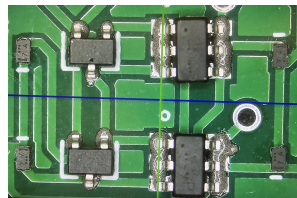
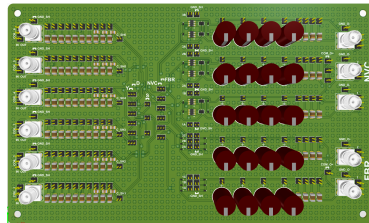


Negative voltage converter for one anti-series pair.

The prototype board



- Custom PCB designed in Altium Designer.
- Every configuration (star, delta, NVC) selected by moving jumpers, on the same hardware.
- Jumper-selectable capacitor bank to set C_m of each coil.
- Assembled entirely by hand, with soldering iron and hot plate.



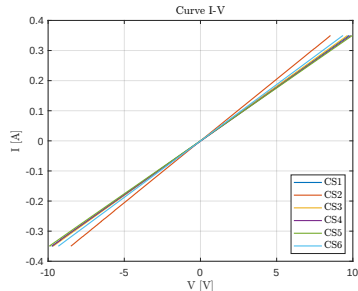
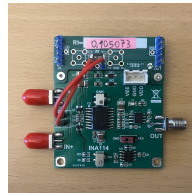
One NVC bridge during hand assembly.



Induction motor (left) coupled to the
MP-VREH (right).

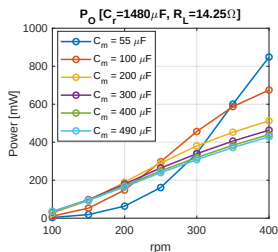
- Induction motor driven by an inverter: rotation speed set precisely over the whole 100–400 rpm range.
- Voltages acquired with a Saleae differential acquisition device.
- Load set by an eight-decade programmable resistor board.

- Six custom sensing boards (INA114AU) measure the drop across a shunt resistor differentially.
- Every shunt characterised with four-wire measurements.
- Each sensor calibrated individually against precision instruments, into its own lookup table.
- This makes power figures trustworthy down to the milliwatt level.

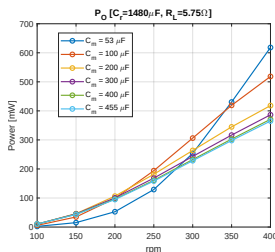


Calibration curves of the six sensors.

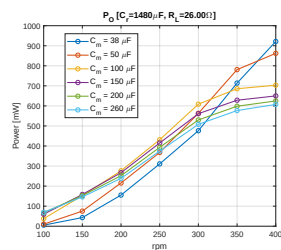
Simulation results



Star

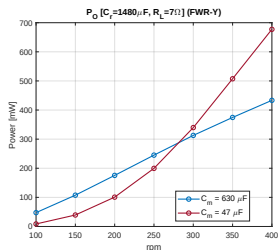


Delta

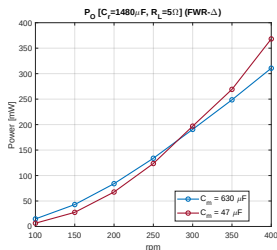


NVC

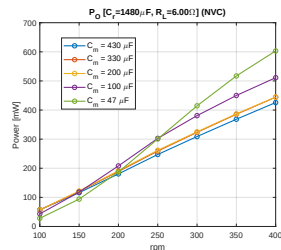
- Output power across the speed range, for several fixed values of C_m .
- At 100 rpm, the **worst case** that sets the lower bound of what the system can deliver, the NVC gives 69.0 mW: about $6\times$ the delta connection (10.7 mW) and nearly double the star one (36.7 mW).
- The NVC is also the most efficient topology at every speed.



Star



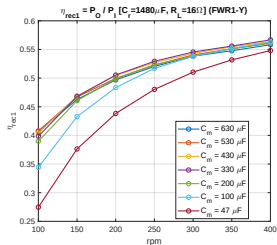
Delta



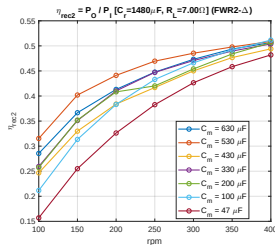
NVC

- At 100 rpm the NVC still wins: 57.7 mW, against 47.6 (star) and 14.9 mW (delta).
- Above roughly 250 rpm the **star connection overtakes the NVC**: 433.3 against 425.7 mW at 400 rpm.
- Simulation alone did not predict this crossover.

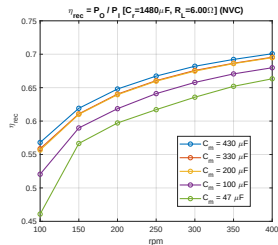
Measured efficiency



Star



Delta



NVC

- The NVC has the **best efficiency at every speed**, up to $\eta_{Rec} \approx 0.70$ at 400 rpm.
- The star connection reaches ≈ 0.55 – 0.57 , the delta connection only ≈ 0.48 – 0.51 : the weakest performer, as expected from its lower line voltage.
- So the two rankings disagree: **efficiency says NVC**, **power says star at high speed**.

Averaged over the whole speed range, the measured power differs from the simulated one by **about 8% for the star connection**, up to **almost 30% for the NVC**.

	Star	Delta	NVC
Mean deviation of P_O from simulation	8%	16%	30%

- The ceramic compensation capacitors **lose part of their capacitance** under the real operating voltage, which grows with speed: a larger nominal C_m is needed to reach the same effective value.
- Tolerances of the real components, and the parasitic resistance of the bridges, add losses the ideal model does not include.
- The sources are not perfectly sinusoidal.

A PMIC conditions the rectified power and delivers a **stable supply** to the load: for example a node whose sensors detect and prevent malfunctions of the system or the vehicle it is installed on.

PMIC (configuration)	V_{in} range (V)	V_{cs} (V)	Star connection
DS-AEM13920_VR	0.120–4.6	0.275	up to 250 rpm
DS-AEM10941_VR	0.05–5	0.38	up to 250 rpm
bq25570	0.1–5.1	0.6	up to 250 rpm
SPV1050 (Boost)	0.15–5.3	0.55–0.58	up to 250 rpm
SPV1050 (Buck-Boost)	0.15–18	2.6–2.8	full range
SPV1040	0.3–5.5	0.8–2	up to 300 rpm

- Delta and NVC stay below about 4 V over the whole range: **compatible with every configuration tested**.
- The star connection climbs to 7 V at 400 rpm: it needs a wide-input converter or an input clamp.

Conclusions

The NVC and the star connection are the two strongest candidates, each suited to a different part of the speed range; the delta connection is consistently the weakest. NVC and delta connect safely to every PMIC tested, while the star connection needs extra protection at high speed: its higher open-circuit voltage can damage the input of the PMIC.

Limitation

The harvester prototype is 3D-printed, and therefore sensitive to vibration, especially at high speed: this limits how repeatable the measurements can be.

Future work

An adaptive compensation network tracking the optimal C_m as speed changes, and a full hardware test of the harvester connected to a real PMIC, to measure the true efficiency from the harvester to the final load.

Thank you for your attention!

Backup: star connection



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rpm	P_O (mW)		η_{Rec}	
	Simulated	Measured	Simulated	Measured
100	36.7	47.6	0.430	0.423
150	94.3	107.5	0.541	0.475
200	165.8	175.6	0.608	0.507
250	240.0	245.1	0.650	0.529
300	308.8	312.7	0.674	0.544
350	372.3	374.6	0.691	0.556
400	430.0	433.3	0.710	0.565

Backup: delta and NVC



Delta

rpm	P_O (mW)		η_{Rec}	
	Sim.	Meas.	Sim.	Meas.
100	10.7	14.9	0.210	0.279
150	43.7	43.2	0.340	0.379
200	95.0	84.2	0.425	0.441
250	158.3	134.1	0.482	0.480
300	228.5	190.8	0.522	0.508
350	298.0	248.7	0.550	0.515
400	365.0	310.7	0.570	0.511

NVC

rpm	P_O (mW)		η_{Rec}	
	Sim.	Meas.	Sim.	Meas.
100	69.0	57.7	0.665	0.568
150	146.5	109.4	0.768	0.619
200	238.9	175.9	0.821	0.648
250	374.8	245.2	0.841	0.668
300	508.0	308.0	0.842	0.682
350	576.8	367.7	0.834	0.693
400	610.0	425.7	0.820	0.701

rpm	Star		Delta		NVC
	V_{OC} (V)	ripple	V_{OC} (V)	ripple	V_{OC} (V)
100	1.98	0.012	1.000	0.022	2.70
150	2.83	0.012	1.460	0.021	3.05
200	3.67	0.011	1.910	0.017	3.20
250	4.51	0.012	2.365	0.017	3.15
300	5.32	0.012	2.825	0.017	3.15
350	6.17	0.011	3.275	0.017	3.15
400	7.04	0.012	3.755	0.017	3.15

The NVC saturates at 3.15 V above 200 rpm, with a ripple an order of magnitude larger than the FWR connections, consistent with its active switching operation.